

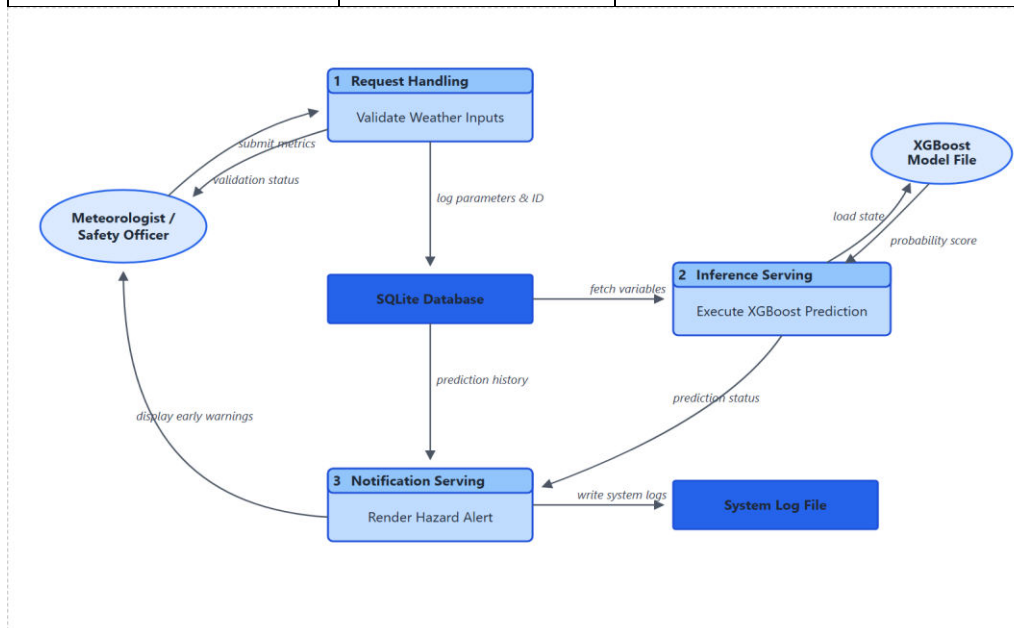
Data Flow Diagram

Date	26 June 2026
Team ID	
Project Name	Rising Waters: A Machine Learning Approach to Flood Prediction
Maximum Marks	2 Marks

Data Flow Diagram

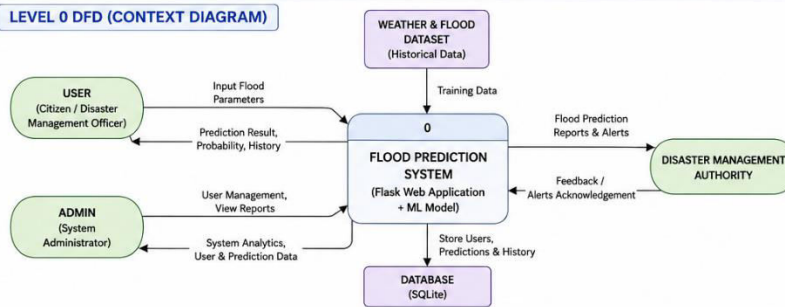
DFD Symbol Legend

Symbol	Name	Description
Oval / Rounded shape	External Entity	A person, organization, or system outside the project that sends or receives data (e.g., Meteorologist, User, Admin).
Rectangle with numbered header	Process	An activity that transforms incoming data into outgoing data (e.g., Validate Request, Compute Inferences, Format Response).
Rectangle (solid fill, no number)	Data Store	A place where data is held for later use (e.g., SQLite Database, Weather Logs database, XGBoost Model pickle).
Labeled arrow	Data Flow	The movement of data between entities, processes, and data stores, labeled with what data is being passed.



DATA FLOW DIAGRAM (DFD) – FLOOD PREDICTION SYSTEM

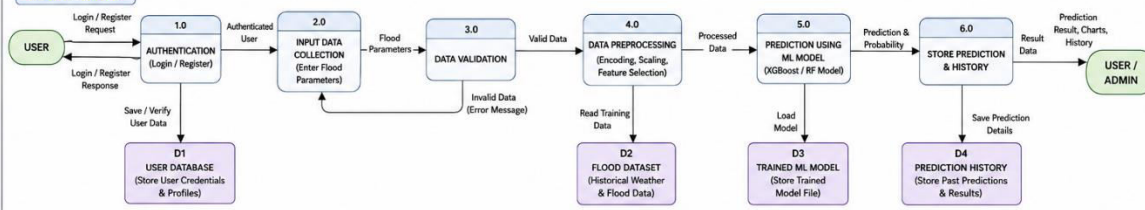
LEVEL 0 DFD (CONTEXT DIAGRAM)



DFD SYMBOL LEGEND

Symbol	Name	Description
	External Entity	A person, organization, or system outside the project that sends or receives data.
	Process	An activity that transforms incoming data into outgoing data.
	Data Store	A place where data is held for later use.
	Data Flow	The movement of data between entities, processes, and data stores.

LEVEL 1 DFD



EXTERNAL ENTITIES

- User (Citizen / Disaster Management Officer)
- Admin (System Administrator)
- Disaster Management Authority
- Weather & Flood Dataset (External Source)

DATA STORES

- D1 – User Database (User credentials, profiles, roles)
- D2 – Flood Dataset (Historical weather & flood data)
- D3 – Trained ML Model (Serialized model file)
- D4 – Prediction History (All past predictions & results)

PROCESSES

- 1.0 Authentication (Login / Register)
- 2.0 Input Data Collection
- 3.0 Data Validation
- 4.0 Data Preprocessing
- 5.0 Prediction Using ML Model
- 6.0 Store Prediction & History
- 7.0 Display Result

DATA FLOWS (EXAMPLES)

- User → System : Login request, flood parameters
- System → User : Prediction result, history, probability
- System → Database : Store users, predictions, history
- System → Authority : Flood prediction reports & alerts
- Dataset → System : Historical training data